

CS 8803/4803 Sociotechnical Security

Semester Project Guidelines

Georgia Tech, Fall 2026

Introduction

Over the course of the semester, you will work in groups of 1–3 people to conduct original research in Sociotechnical Security. This document will discuss the expectations and major checkpoints for this assignment, as well as provide you with rough guidelines on how to conduct open-ended research in this area.

This project and its constituent parts are worth 50% of your grade in the course. You **may not** use generative AI to complete any part of the project, except for checking spelling and grammar.

Forming a group

Projects should be conducted in teams of 1–3 students. You can choose to work with whoever you like in the class – there are no pre-assigned groups. If you would like to work with someone but are having trouble finding a group, talk to Veronica and she will facilitate.

Project options

You can choose between two different project options in the course, an open-ended option or a platform-centered option. The expectation of effort for both options is the same—the latter simply provides more structure while the former is more open-ended. The project milestones, submission requirements, and grading criteria are the same for both options. Both are detailed below.

Option 1: Open-ended research project

In this option, students will come up with their own, original research idea and execute on the idea. The idea must be thematically related to the material in the course in some way – students choosing this option will be asked to justify its relevance to the course in the project intention and midpoint check-in document.

Students will be expected to do the following:

- Come up with a research idea area relevant to the topics presented in the class. We will not be able to cover all possible topics before you commit to a project so check the course website for a list of topics since you may be interested in a topic we won't cover until later in the course.
- Read and synthesize related papers to the topic area
- Create a research plan that details the hypotheses, data, tests, systems, designs, or other research effort required to complete the project
- Identify how to define whether the project was successful (this should be more specific than simply "I finished the project.")
- Execute the research plan

Option 2: Platform harms measurement project

In this option, students will conduct a platform measurement project. This project involves the selection of a specific type of harm and a platform, the creation of an automated detector for identifying that harm, and a thorough analysis of the platform guided by specific research questions.

Students will be expected to do the following:

- Identify a social platform of interest (broadly defined) and a specific harm that occurs on the platform. Selection of the platform should take into consideration availability of data or ease of data access.

- Read and synthesize related papers that have studied this platform in the past.
- Create a research plan that details the research questions you are seeking to answer, the data you are planning to collect, the hypotheses that you plan to test, and the analyses that you plan to conduct to answer your research questions.
- Design and build an automated detection mechanism to identify the harm you are interested in studying. You can use any tools you like, but it should be specifically evaluated with your platform in mind.
- Identify how to define whether the project was successful (this should be more specific than simply “I finished the project.”)
- Execute the research plan.

If you are already conducting research in this area with a faculty member on campus and wish to use your ongoing research as a part of your course requirements, please discuss this with me in advance of submitting your project proposal.

Milestones

There are three main milestones for this project: a project proposal, a project midpoint check-in document, and a final presentation and document. Each is detailed below:

Project proposal (10%), due Sep. 14 2:00 PM

The project proposal is 10% of your project grade. It commits you to a group in the class and sets the main direction for your project. With your group, you will write a 1-2 page document that details:

1. The names of the people in your group
2. Which of the two project options you have selected
3. The motivation for your project
4. The research questions that will guide your project
5. Your research or measurement plan
6. Anticipated challenges or risks
7. Expected outcomes & how you will measure success

Detailed proposal instructions will be linked on the course website. After you submit your proposal, your group will meet with Veronica to receive feedback on your intended direction. You will be graded on the timeliness and completeness of your submission, including attending the meeting with Veronica.

Project midpoint check-in (10%), due Oct. 14, 2:00 PM

The project midpoint report is 10% of your project grade. The goal of this report is for you to describe your current status on the project and get help from Veronica early so your project can be successful. With your group, you will write a 2-3 page document that details:

1. An introduction which frames the problem you are studying
2. A related work section that details prior work relevant to your topic
3. Your research plan and current status for executing on that research plan
4. Any challenges you have encountered so far
5. An expected timeline for how you will complete the project.

Detailed midpoint report instructions will be linked on the course website. After you submit your report, your group will meet with Veronica to receive feedback on your intended direction. You will be graded on the timeliness and completeness of your submission, including attending the meeting with Veronica.

Project final presentation and report (30%)

Students will be required to put together a 15-20 minute presentation on their project, to be presented during the final class date (December 7th). The presentation will be followed by a question and answer (Q&A) session of approximately 5-10 minutes with the class. The Q&A may be longer or shorter depending on how many groups need to present. This presentation is worth 15% of your overall grade. You will be graded based on clarity of presentation, the rigor of your research project, and how well you handle audience questions. A more concrete rubric will be shared closer to the week of presentations.

You will also prepare a 5-6 page document that details the progress you made over the course of the project. This report is worth 15% of your overall grade. The details of this document will vary from team to team, but the sections that must be included are:

- An introduction which frames the problem you are studying and presents the topline findings of your research
- A related works section which details the prior work related to your topic and how your work relates and differs from these works.
- A methodology section which discusses how you set out to answer your research question
- A results section which documents your findings
- A discussion and future work section which puts your findings in the context of prior work, and details what future work might exist in this area.

The final report will be graded on the timeliness and completeness of your submission, the quality of your submission, and how well you frame your work in the context of prior work.

Conclusion

Ultimately, this project should help you get exposed to conducting original research in Sociotechnical Security. Veronica is always happy to discuss your ideas and help you to frame and shape your work as necessary. Have fun!

Acknowledgements: This assignment was inspired by a similar assignment in UCSD's Sociotechnical Cybersecurity and NYU's Trust and Safety courses.